

Multi-View VLM Verification for Semantic Object Extraction from a Terrestrial Laser Scan Toward TOSM-Based Robot Environment Models

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Abstract—Semantic environment models such as the Triplet Ontological Semantic Model (TOSM) give indoor mobile robots the symbolic, explicit, and implicit knowledge needed for task-level autonomy, but populating them for an existing facility is still manual work. This paper presents an automated pipeline that extracts TOSM semantic objects from a single terrestrial laser scan and emits them as a VDA 5050-style JSON message. A deterministic stage segments the colored point cloud and fills the explicit model (pose, dimensions, color) with a provisional class from an open-vocabulary 3D classifier; a vision–language model (VLM) stage then verifies each class from multi-view renders under forced tool-use and infers the implicit affordance model. We show that the obvious multi-view strategy — feeding all views into one prompt (early fusion) — *degrades* accuracy on sparse point renders, and propose per-view classification with confidence-weighted voting (late fusion) instead. On a 68.5M-point scan of a robot showroom (57 objects), graded both by a render-based audit and by an independent on-site inventory, late fusion is the strongest multi-view strategy (74% vs. 67% lenient accuracy over early fusion), although a single canonical view remains the most accurate configuration overall. The robust benefit of the multi-view stage is a free reliability signal: label votes unanimous across views are 94–95% correct against the inventory in three independent runs, versus 30% for split votes — gating validated when the inventory overturned labels that had fooled both the VLM and the human audit. Full-scene verification costs \$2.6; prompts, settings, and audits are released.

Index Terms—multi-view object recognition, vision–language model, semantic mapping, terrestrial laser scanning, TOSM, VDA 5050

I. INTRODUCTION

Task-level autonomy for indoor mobile robots requires more than an occupancy grid: the robot must know *what* occupies the space, *where*, and *how* it can be interacted with. The Triplet Ontological Semantic Model (TOSM) [1] organizes this knowledge into a *symbolic* model (class, name, identity), an *explicit* model (pose, size, color), and an *implicit* model (affordances such as whether an object is a reliable landmark or can be displaced). TOSM-style databases drive navigation frameworks and robot fleets, and their content maps naturally onto the JSON conventions of VDA 5050 [2], the emerging AGV-to-master-control interface standard.

In practice, the semantic-object database for a new facility is authored by hand. Terrestrial laser scanners (TLS) such as

the Leica BLK360 capture a dense, colored as-built model in minutes, and scan-to-BIM research [3] recovers structural geometry, but object-level semantics remain the gap: open-vocabulary 3D classifiers such as Uni3D [4] are biased toward industrial structural classes — in our experiments office chairs were labeled *stair* or *machine* and displays *ladder* — and no point-cloud classifier produces the affordance-level knowledge TOSM needs.

Vision–language models (VLMs) offer a complementary signal: rendered views of a segmented object carry texture and context that a point-cloud embedding discards. We render each object from four azimuths and study how a VLM should consume them. The naive strategy — all views in one prompt (*early fusion*) — *hurts*: sparse oblique point renders invite over-confident misreadings that contaminate the joint judgment. Classifying each view independently and voting by confidence (*late fusion*), the analogue of MVCNN-style aggregation [5], recovers most of the lost reliability and exposes inter-view disagreement as an uncertainty signal. The contributions are:

- an automated TLS-to-TOSM pipeline whose output is a VDA 5050-style `semanticObjects` message consumable by a fleet environment model;
- a schema-forced VLM verification protocol with per-view late fusion, plus prompt guardrails (size sanity, sparse-render caution, panel disambiguation) for point-render inputs;
- a six-configuration study on a real 68.5M-point BLK360 scan (57 objects), graded by a render-based audit and re-scored against an independent on-site inventory with repeat runs, showing that early fusion degrades accuracy, that late fusion is the strongest multi-view strategy while a single canonical view stays most accurate overall, and that vote unanimity yields 94–95%-precise gating.

II. RELATED WORK

Semantic environment models. TOSM [1] extends ontology-based robot knowledge bases [6] with the symbolic/explicit/implicit triplet; its records subsume the object fields of VDA 5050 [2] messages, which we adopt as the output container; the taxonomy also underpins a knowledge-grounded 3D scene graph for LLM querying [7], whose

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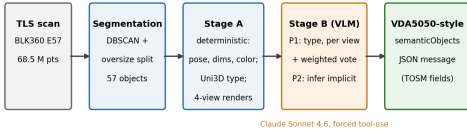


Fig. 1. Scan-to-semantics pipeline. Stage A deterministically fills the explicit TOSM model plus a provisional class; Stage B (Claude Sonnet 4.6, forced tool-use) verifies the symbolic model per view with weighted late fusion and infers the implicit model.

records we populate automatically. **Scan-to-model.** As-built modeling from TLS targets structural elements [3]; object semantics are left to learned 3D segmentation [8] or open-vocabulary encoders such as Uni3D [4], which inherit training-corpus biases. **Multi-view recognition.** Aggregating rendered views is a classic route to 3D shape recognition (MVCNN [5]); we revisit early-vs-late fusion for VLM prompts over *sparse point* renders. **VLMs in robotics.** Recent systems use VLMs for open-set scene graphs [9] and language-driven maps [10]; we use one in a narrower, auditable role: verifying one object at a time against measured geometry.

III. PIPELINE

A. Stage A: Deterministic Explicit Model

The input is a single registered BLK360 scan (E57, colored), voxel- downsampled at 3cm. Structural planes are removed by seeded RANSAC (300 triples per plane, 5cm inlier threshold, SVD refit; only near-horizontal $|n_z| \geq 0.85$ or near-vertical $|n_z| \leq 0.2$ planes are removed; stop after ten planes, below 2,000 remaining points, or when the best plane holds $<4\%$ of them or is slanted). Objects are then segmented by DBSCAN [11] ($\epsilon = 0.30$ m, $\text{minPts} = 100$), which yields 30 clusters on our scene; every cluster whose footprint exceeds 2.5 m is re-clustered at $\epsilon = 0.20$ m (split accepted only if it yields ≥ 2 sub-clusters of ≥ 100 points), raising the count to the 57 objects evaluated in Sec. IV-A with no further filtering. Segmentation stays geometric because ScanNet-trained segmenters did not transfer to this scene (SPFormer labeled only 51% of points, PTv3 called 46% *wall* after wall removal). For each object Stage A computes the explicit TOSM model directly from the points: bounding-box center (`poseX/Y`, `properties.poseZ`), PCA-based yaw (`poseTheta`), yaw-aligned extents (dimensions), and a mean RGB color name. Uni3D provides a *provisional* symbolic class, and four azimuth renders (0/90/180/270°, elevation 25°, 512×512, point size 8, red 3D box) are exported; smaller point sizes produced speckle-like views that misled both the VLM and the human auditors (Sec. IV-B). Stage-A fields are measured, not generated; Stage B never modifies them.

B. Stage B: VLM Verification with Late Fusion

Stage B calls a commercial VLM (Claude Sonnet 4.6, API model `claude-sonnet-4-6`, 512 output tokens, default sampling temperature) with *forced tool-use*: the model must answer by calling a declared JSON tool, so every response is

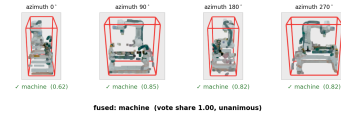


Fig. 2. Late fusion on one object (*machine_004*). Each of the four azimuth renders is classified independently; here all views agree, yielding the unanimous, confidence-1.0 case that both the render audit and the inventory find 94–95% reliable (Sec. IV-B).

P1 system (condensed). Verifier for an indoor mobile robot. Input: ONE render of a segmented 3D object (sparse laser points, red 3D box), its classifier label `uni3d_type`, optional candidates. *Decision rule:* (1) KEEP a correct `uni3d_type`; (2) else replace — prefer a fitting candidate, otherwise the most accurate common-noun label; never ‘unknown’. *Size sanity:* measured dimensions are a HARD constraint; an oversize cluster is merged objects — label the dominant structure or ‘clutter’. *Sparse-render caution:* never claim parts not outlined by points. *Panel rule:* thin flat objects are keyboard / display / shelf — never ladder. *Chair gate:* visible seat pan 0.3–0.6 m plus backrest. If uncertain, keep a size-compatible label. Answer ONLY via `report_symbolic`.
P2 system (condensed). From views and confirmed type, report `isKeyObject` (fixed, distinctive landmark), `isMovable`, `isOpen/canBeOpen` (doors only, else null) ONLY via `report_implicit`.

Fig. 3. Condensed Stage-B prompt design (full text in the released code). Forced tool-use makes every response schema-valid JSON; measured geometry is read-only context.

schema-valid by construction. Fig. 3 condenses the prompt design; the full system prompts, tool schemas, and configuration files are released with the code¹.

Prompt 1 (symbolic), once per view. The VLM sees ONE render, the measured dimensions, the provisional class, and an optional candidate vocabulary — not generated per object but one operator-supplied scene-level list (*chair, TV, monitor, Mobile Robot, desk* here) passed unchanged to every call. The decision rule is three-way: keep a correct label; if wrong, prefer a fitting candidate; else emit an open-vocabulary label. Guardrails address point-render failure modes (Fig. 3): dimensions as a hard constraint, oversize clusters as merged objects, thin-panel disambiguation, no parts not outlined by points, and a keep-or-clutter fallback.

Late fusion. The four per-view reports are aggregated by confidence-weighted voting: each label ℓ accumulates the confidences of the views that voted for it, $s(\ell) = \sum_{v: \ell_v = \ell} c_v$; the top-scoring label wins, and the final confidence is its share $s(\ell^*) / \sum_{\ell} s(\ell)$. The per-view votes are stored in the record (`properties.viewVotes`), so a downstream consumer can distinguish a 4/4 unanimous label from a 2/4 split — a free uncertainty signal. Fig. 2 shows a unanimous case (fused confidence 1.0). Conversely, a single contaminated view is simply outvoted (a tall-backed chair whose frontal view read *Mobile Robot* fused to *chair* 3:1), whereas inside one early-fusion prompt the same view can drag the joint judgment toward the wrong label.

Prompt 2 (implicit), once per object. Given the fused type and all views, the VLM reports the TOSM implicit model: `isKeyObject`, `isMovable`, and (doors only) `isOpen/canBeOpen`.

¹<https://github.com/kkk3449/tls-smf>

TABLE I
TOSM $\rightarrow$ VDA 5050-STYLE JSON FIELD MAPPING

TOSM model	element	JSON field
symbolic	class	type
	name / identity	name, id
explicit	position, height	poseX, poseY, properties.poseZ
	orientation, size	poseTheta, dimensions{l, w, h}
	color	color
implicit	landmark / movability	properties.isKeyObject,
	door state	isMovable, isOpen, canBeOpen
–	VLM audit trail	properties.viewVotes,
		symbolicReason, confidence

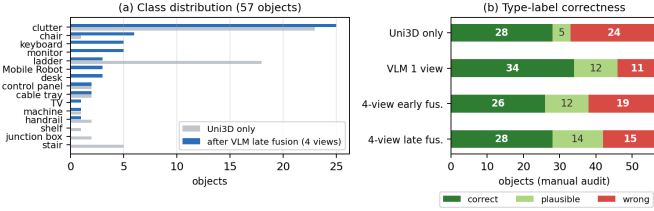


Fig. 4. Effect of VLM verification on the 57-object scene. (a) Class distribution, classifier-only vs. after late fusion: implausible *ladder/stair* populations are redistributed to displays and furniture (the keyboard outputs were later rejected by the on-site inventory, Sec. IV-B). (b) Render audit: early fusion degrades single-view accuracy; late fusion is the strongest multi-view strategy (15 wrong labels vs. 19 for early fusion).

C. TOSM to VDA 5050-Style Message

The completed records are serialized as one `semanticObjects` JSON message with a VDA 5050-style header (Table I); TOSM fields with no standard slot travel in the extensible `properties` map, so consumers that only know the base schema can still read it.

IV. EXPERIMENTS

A. Setup, Audit Protocol, and Reference Labels

We evaluate on a single BLK360 scan of an indoor robot showroom (68.5M colored points, 57 segmented objects) with Claude Sonnet 4.6. The six configurations (Table II) form a 2×2 design over view count (1 vs. 4, early fusion) and render/prompt generation (sparse renders + base prompt vs. dense renders + guardrail prompt), plus a sixth that differs from four-view early fusion (dense) *only* in the fusion rule: render density and guardrails were upgraded together and are not separable here, whereas the fusion contrast is isolated. Metered cost: late fusion \$2.63 per scene (285 calls, \$0.046/object); early fusion \$1.27; single view \$0.87.

Labels are graded by two independent references. (i) *Render audit*: two of the authors audited all 57 objects against the dense four-view renders and the measured dimensions, rating every configuration’s final label *correct*, *plausible* (consistent with the views but not verifiable from renders alone), or *wrong*. (ii) *On-site inventory*: the facility operator then assigned every object a ground-truth type by physical inspection, without reference to the render audit (43 confirmed, 12 plausible, 2 unknown; 7 clusters are ceiling-soffit structure remnants). Inventory scoring is strict — a label counts only if it equals

	Symbolic	type	control panel
		name	control_panel_016
		verified	true (VLM 4-view)
	Explicit	pose	x 7.74 y 4.20
			z 0.88 m (map)
			-3.07 rad
		theta	1.93 x 0.60 x 2.51
		dims	brown
		color	1.00
		conf.	
	Symbolic	type	chair
		name	chair_005
		verified	true (VLM 4-view)
	Explicit	pose	x 3.89 y -1.54
			z -0.50 m (map)
			-2.61 rad
		theta	1.28 x 0.58 x 1.00 m
		dims	black
		color	1.00
		conf.	
	Implicit	keyObject	false
		movable	true

Fig. 5. Completed TOSM semantic-object records (dense render + symbolic / explicit / implicit fields) as serialized into the VDA 5050-style message. (a) A control panel, unanimously verified and marked as a fixed key landmark. (b) An office chair corrected from *stair*, movable and not a landmark.

the inventory type, so a conservative *clutter* on a real chair is wrong — on all 57 objects and on the 36 confirmed non-structure objects. Audit-vs-inventory agreement is 79–95% over the five VLM configurations (Cohen’s κ 0.53–0.89; 89%, $\kappa = 0.79$ for late fusion): the audit is a faithful but imperfect proxy whose optimistic calls are confined to *handrail* labels on soffit remnants and whose single pessimistic call concerns an object the inventory itself marks unknown.

B. Multi-View Study

Table II and Fig. 4(b) summarize the audit. Three findings stand out.

(1) *Early fusion hurts*. Feeding four sparse views into one prompt dropped strict accuracy from 60% (single view) to 49%: extra oblique speckle views encouraged over-committed readings, e.g. two electrical cabinets confidently relabeled *chair*. Denser renders and the prompt guardrails did not close the gap (46% vs. 51%). The inventory confirms the direction: four-view early fusion is the worst VLM configuration under both references.

(2) *Late fusion is the strongest multi-view strategy, but not the best configuration*. Judging each view independently and voting beats early fusion on every measure (15 vs. 19 wrong labels, 74% vs. 67% lenient; 34 vs. 30 inventory matches): a contaminated view is outvoted. It does not overtake the best single canonical view (60%/81% audit; 44/57 inventory matches, 77%, 95% Wilson CI 65–86%, vs. 60%, CI 47–71% for late fusion): on sparse point clouds extra views add noise faster than evidence, and the vote spends much of its gain on caution. Its larger *plausible* bucket — split votes often land on *clutter* — counts as wrong under the strict inventory, which is why late fusion exceeds the classifier by only one object there (34 vs. 33). We therefore claim late fusion as the safer multi-view rule, not as an accuracy gain over a well-chosen single view; the measurable value of the multi-view stage is the reliability signal below.

(3) *Vote unanimity is a free precision signal*. 17 of 57 objects received identical labels from all four views; 16 of those (94%) were audited correct, and the same 16 match the inventory, versus 30% (audit) or 18/40 (inventory) among split-vote objects. Gating on the stored `viewVotes` thus gives a downstream consumer a calibrated route-to-human-review policy that per-call confidence alone did not provide.

TABLE II

FINAL TYPE LABELS (57 OBJECTS) UNDER BOTH REFERENCES. RENDER AUDIT: C/P/W = CORRECT / PLAUSIBLE / WRONG, STRICT = C, LENIENT = C+P. INVENTORY: STRICT MATCHES ON ALL 57 OBJECTS AND ON THE 36 CONFIRMED NON-STRUCTURE OBJECTS.

Configuration	Render audit						Inventory	
	C	P	W	strict	lenient		57	36
Uni3D only (no VLM)	28	5	24	49%	58%		33	28
1 view, base prompt, sparse	34	12	11	60%	81%		44	33
4 views early fus., sparse	28	10	19	49%	67%		34	27
1 view, guardrails, dense	29	11	17	51%	70%		32	29
4 views early fus., dense	26	12	19	46%	67%		30	23
4 views late fus., dense	28	14	15	49%	74%		34	28

Repeatability. Because the VLM samples stochastically, we later re-ran the late-fusion protocol twice on the 50 non-structure objects. Labels agreed between the re-runs on 82% of objects; accuracy on the 36 confirmed objects was 28, 27, and 27 across the three runs (± 1), unanimity precision 16/17, 20/21, and 19/20 (94–95%), split-vote accuracy 45%, 28%, and 30%. Rankings are stable to sampling, but the single-scene 57-object sample yields intervals of roughly ± 12 points, so all comparative claims are scoped to this scene.

Render density cut both ways, and produced the study’s starkest warning: dense renders made four dotted-grid panels look so convincingly like keyboards (per-view confidence up to 0.97) that the label passed our render-based audit — only the inventory revealed them to be ceiling-soffit remnants, and Table II grades every such label wrong. A label that fools both the VLM and a human at the same renders was still caught by vote splits (these objects never reached unanimity), the strongest argument for the gating above; the incident also motivated a structure-remnant filter now applied in Stage A.

C. Implicit Model and Completed Records

On the late-fusion output, Prompt 2 marked 6 of 57 objects as key objects (control panels, a large machine, mobile robots, a wall display) and 50 as movable. Fig. 5 shows two completed TOSM records as stored in the output message, including the audit trail (per-view votes, fused confidence, one-sentence reason). Mean fused confidence was 0.71; the 17 unanimous objects carry 1.0 by construction, matching the observed 94% unanimity precision. By design the records are a reviewable draft, not ground truth: the model stays user-editable and every amendment is logged. An owner review exercised this loop on the implicit model, correcting two movability entries (a wheeled machine judged fixed, a wall-mounted display judged movable) at negligible cost.

D. Discussion and Limitations

The division of labor is the main design point: everything measurable stays deterministic and VLM-immutable, so hallucination can corrupt neither poses nor dimensions, and the VLM contributes exactly the judgments geometry cannot make — naming and affordances — in an auditable form (per-view votes, forced-tool JSON, stored reasons). All 741 tool

calls across the five VLM configurations returned schema-valid JSON. Remaining errors concentrate where the render itself is ambiguous (texture-poor slabs, merged clusters, sparse strips), pointing to upstream fixes — finer segmentation, mesh renders — not to the fusion protocol. Absolute strict accuracy (46–60% audit, 53–77% inventory) reflects a deliberately hard test bed: a cluttered showroom scanned once, without cleanup. Limits: a single scene and 57 objects; confounded render-density and prompt factors; an inventory compiled by one operator. Multi-scene evaluation with per-factor ablation is future work.

V. CONCLUSION

We presented an automated pipeline that turns a single terrestrial laser scan into TOSM semantic-object records delivered as a VDA 5050-style JSON message, using a VLM in a tightly scoped, schema-forced role. A six-configuration study, graded by a render audit and an independent on-site inventory, showed that the intuitive all-views-at-once prompt degrades verification on sparse point renders, whereas per-view classification with confidence-weighted late fusion is the strongest multi-view strategy (wrong labels 24→15, 74% lenient), although a single canonical view remains the most accurate configuration on this scene. The robust benefit of the multi-view stage is its vote signal: unanimity gives a 94–95%-precise gate across three independent runs at \$2.6 per scene. Future work includes mesh renders, finer segmentation, multi-scene evaluation, and coupling the records to place modeling and a live digital twin.

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